

Brown Dwarfs as Laboratories for Giant Exoplanets: Exploring Disequilibrium Chemistry in Substellar Atmospheres

Cold brown dwarfs ($250 \leq T \leq 500$ K) are effective analogs for directly imaged giant exoplanets because they share similar atmospheric physics but are more numerous and easier to observe at high signal-to-noise. As isolated objects, they provide a controlled laboratory for studying atmospheric processes that are difficult to constrain in exoplanets. Atmospheric models found that seven cold brown dwarfs exhibit disequilibrium CO abundances driven by strong vertical mixing, parameterized by the eddy diffusion coefficient (K_{zz}). These results imply that vertical mixing plays a critical role in shaping observable chemistry in cold atmospheres and must be included in exoplanet models. However, the same models predict detectable phosphine (PH_3), which has not been observed, indicating a discrepancy between model predictions and observations. We propose to analyze 14 cold brown dwarfs observed with JWST program GO 8441, expanding beyond the original sample. Using chemical quenching theory and varying K_{zz} , we will test whether key trends, enhanced CO at low temperatures and predicted PH_3 , persist across a larger population, and determine whether PH_3 is absent or below detection limits. This work will constrain vertical mixing efficiencies and disequilibrium chemistry in cold substellar atmospheres, improving models used to interpret JWST spectra of directly imaged giant exoplanets. These unique results will provide important insight into the atmospheres and evolution of directly imaged giant exoplanets.

Missing Molecules in Cold Brown Dwarfs: Constraining Disequilibrium Chemistry and Vertical Mixing

Brown dwarfs exhibit complex disequilibrium chemistry driven by atmospheric variability and vertical transport processes. Although only a limited number of studies have focused on disequilibrium chemistry in isolated brown dwarfs, these works highlight the importance of understanding such atmospheres, which serve as key analogues for directly imaged giant exoplanets. Miles et al. (2020) find that cold brown dwarfs exhibit strong vertical mixing and significant CO disequilibrium chemistry. Similarly, Morley et al. (2018) show that while simple molecules such as CH₄ are readily detected in the coldest brown dwarfs, predicted complex species like PH₃ are notably absent, pointing to incomplete or poorly understood phosphorus chemistry in cold substellar atmospheres. Lodders & Fegley (2002) further predict that brown dwarf atmospheres are dominated by a small set of simple molecules (e.g., CO, CH₄, H₂O, NH₃), such that the presence of additional species, or their absence when expected under disequilibrium, serves as a key diagnostic of vertical mixing and chemical kinetics. Because vertical mixing can drive the production and transport of complex molecules, studying these processes provides critical insight into the atmospheric evolution of substellar objects.

However, several expected complex species, such as PH₃ remained unexplained in the atmospheres of these objects, as shown in Miles et al. 2020 and references therein. With JWST now providing high-precision spectra of cold substellar atmospheres, the mismatch between observed and predicted disequilibrium species has become a limiting factor in interpreting these data. Resolving this gap is essential for translating JWST observations into robust constraints on atmospheric chemistry, mixing, and planetary/brown dwarf evolution.

We propose a theoretical investigation of 14 brown dwarfs observed in JWST program GO 8441 to model disequilibrium chemistry and vertical mixing in isolated substellar atmospheres. The James Webb Space Telescope provides the sensitivity and infrared wavelength coverage necessary to detect key molecular features in cold brown dwarfs, particularly in the near- and mid-infrared where disequilibrium species are expected to emerge. Its demonstrated ability to obtain high signal-to-noise spectra of cold substellar objects makes it uniquely suited to test predictions of vertical mixing and chemical kinetics. This work will deliver two primary outcomes: (1) improved understanding of atmospheric evolution in cold substellar objects, including directly imaged exoplanets, by quantifying how vertical mixing and chemical kinetics shape observable molecular abundances; and (2) the public release of validated chemical quenching and mixing models, along with derived atmospheric parameters for the GO 8441 sample, enabling future studies of substellar atmospheres.

Missing Disequilibrium Chemistry in Cold Brown Dwarf Atmospheres

Cold brown dwarfs exhibit enhanced abundances of CO that deviate significantly from expectations under chemical equilibrium, where CO should be efficiently converted into CH₄ at low temperatures. The persistence of CO therefore implies strong vertical mixing and chemical quenching. If such mixing is sufficient to quench CO, it should also influence other species with comparable or longer chemical timescales. Phosphine (PH₃) provides a key test case: it is

expected to form in deep, hot atmospheric layers and should be absent in observable regions under equilibrium conditions, yet it ought to be dredged upward and preserved by the same mixing processes responsible for enhanced CO. **However, observations show that PH₃ detections are weak, absent, or inconsistent, revealing a fundamental discrepancy.** This inconsistency suggests missing physics in current models, potentially including incomplete or inaccurate chemical kinetics, misestimated vertical mixing strengths (K_{zz}), or additional atmospheric processes that have not yet been accounted for. Previous studies (e.g., Miles et al. 2020; Morley et al. 2018; Leggett et al. 2012) highlight this unresolved issue.

Addressing this problem is critical for developing a self-consistent framework that links vertical mixing and chemical kinetics, ultimately enabling robust interpretation of atmospheric compositions. Such a framework is essential for translating observed spectra into constraints on atmospheric structure, evolution, and formation pathways for both brown dwarfs and directly imaged exoplanets. The key missing component is a self-consistent treatment of disequilibrium chemistry that captures the interplay between vertical mixing (K_{zz}), chemical reaction networks, and quenching processes in determining the abundances of complex species such as PH₃. This program will address this gap by applying chemical quenching theory and mixing models to an expanded sample of cold, isolated brown dwarfs observed with JWST, testing whether predicted species should be observable under realistic conditions and determining whether the absence of complex species arises from incorrect chemical kinetics, misestimated mixing strengths, or previously unaccounted atmospheric processes.

Testing Disequilibrium Chemistry with Chemical Quenching and Vertical Mixing in Brown Dwarf Atmospheres

In this archival program, we target a sample of 14 cold brown dwarfs (250–500 K) observed with JWST using spectroscopic data. These objects provide a well-characterized benchmark set for investigating disequilibrium chemistry and vertical mixing in cold substellar atmospheres. JWST’s infrared sensitivity and broad spectral coverage uniquely enable the detection of weak molecular features, such as PH₃, in these low-temperature regimes. This capability is essential for testing predictions of disequilibrium chemistry and assessing the role of vertical transport in shaping atmospheric composition.

We propose to model this sample using a 1D radiative–convective framework. Atmospheric models will be generated with PICASO, parameterized by effective temperature, surface gravity, metallicity, and C/O ratio for each object. For each case, we will compute spectra under both chemical equilibrium and disequilibrium conditions while varying the eddy diffusion coefficient (K_{zz}) to quantify the impact of vertical mixing. Quench levels for key species (e.g., CO and PH₃) will be identified within each atmospheric profile (Figure 1), and synthetic spectra will be directly compared to JWST observations to evaluate model fidelity and constrain mixing efficiencies.

The primary product of this program is a self-consistent framework linking vertical mixing, chemical quenching, and observable spectral signatures in cold brown dwarf

atmospheres. By reconciling discrepancies between CO and PH₃ predictions and observations, this work transforms an observational inconsistency into a physically interpretable diagnostic of atmospheric processes. The models and chemical framework will be made public.

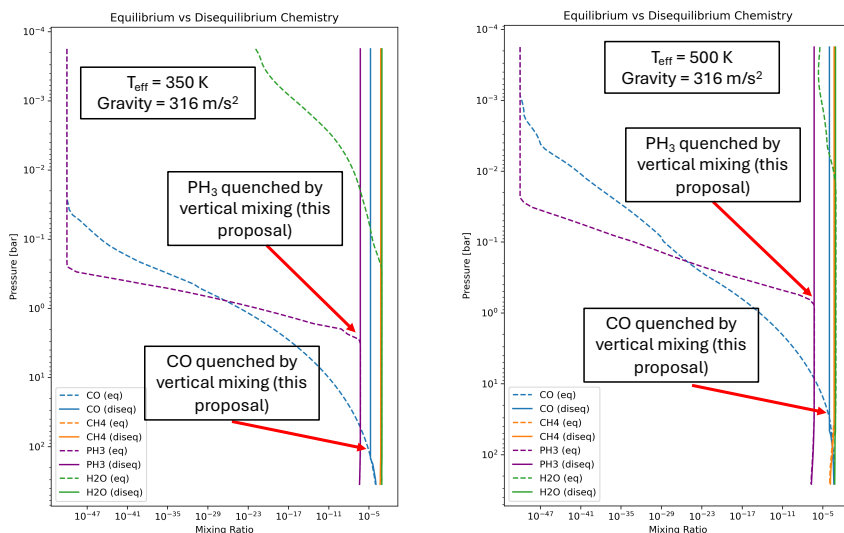


Figure 1. CO is depleted relative to equilibrium, while PH₃ is quenched and maintained at deep-layer abundances. Equilibrium (dashed) and disequilibrium (solid) mixing ratios for CO, CH₄,

PH₃, and H₂O versus pressure for two cold brown dwarfs ($T_{\text{eff}} = 350$ K, left; 500 K, right; $g = 316 \text{ m s}^{-2}$). Disequilibrium models include vertical mixing (K_{zz}), which quenches abundances at depth. **The predicted PH₃, absent in current observations, highlights the impact of vertical mixing and motivates this study.**

Advancing the Interpretations of JWST Spectra Through Disequilibrium Chemistry Constraints

This study will establish robust, quantitative constraints on vertical mixing (K_{zz}) and disequilibrium chemistry in cold brown dwarf atmospheres, directly resolving persistent discrepancies between atmospheric models and JWST observations, such as the absence of predicted species like PH₃. By systematically applying these constraints across an expanded JWST sample, this work will significantly refine atmospheric retrieval frameworks and improve the accuracy of infrared spectral interpretation, advancing our ability to characterize the composition, structure, and physical processes of cold substellar atmospheres.

The rapid expansion of high-precision infrared spectroscopy from JWST makes this program both timely and necessary, as current models are not keeping pace with the quality and diversity of new observations, particularly in the $250\text{--}500$ K regime where complex chemistry and vertical transport dominate. This program directly aligns with JWST's "Other Worlds" science goals by advancing our understanding of atmospheric composition and evolution in substellar objects, which are intrinsically linked to planet formation processes and the characterization of exoplanetary systems. By producing physically validated disequilibrium chemistry models, publicly available analysis tools, and benchmark spectral interpretations, this study will increase the legacy value of JWST by enabling broader community use of its data and supporting cross-disciplinary research in exoplanet atmospheres, brown dwarf physics, and planetary system formation.

Bibliography:

Miles, B. E., Skemer, A. J. I., Morley, C. V., Marley, M. S., Fortney, J. J., Allers, K. N., Faherty, J. K., Geballe, T. R., Visscher, C., Schneider, A. C., Lupu, R., Freedman, R. S., & Bjoraker, G. L. 2020, *AJ*, 160, 63. <https://doi.org/10.3847/1538-3881/ab9114>

Morley, C. V., Skemer, A. J., Allers, K. N., Marley, M. S., Faherty, J. K., Visscher, C., Beiler, S. A., Miles, B. E., Lupu, R., Freedman, R. S., Fortney, J. J., Geballe, T. R., & Bjoraker, G. L. 2018, *ApJ*, 858, 97. <https://doi.org/10.3847/1538-4357/aabe8b>

Lodders, K., & Fegley, B. 2002, *Icarus*, 155, 393–424. <https://doi.org/10.1006/icar.2001.6740>